

Recall: Definition of a Vector space

Definition. A **vector space** over a field $\mathbb{F}$ is a non-empty set V of vectors, together with an operation of vector addition and an operation of multiplication by scalars from $\mathbb{F}$, for which the following properties hold.

1. **Closure under (vector) addition.** If $\vec{u}, \vec{v}$ are in V , then $\vec{u} + \vec{v}$ is in V .
2. **Associativity of (vector) addition.** For all $\vec{u}, \vec{v}, \vec{w}$ in V we have $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$.
3. **Commutativity of (vector) addition.** For all $\vec{u}, \vec{v}$ in V we have $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
4. **Zero vector.** There exists an element $\vec{0}$ of V such that for all $\vec{v} \in V$ we have $\vec{v} + \vec{0} = \vec{v}$.
5. **Negatives.** For each element $\vec{v}$ of V there is an element $\vec{w}$ of V such that $\vec{v} + \vec{w} = \vec{0}$. (We usually denote this element by $-\vec{v}$.)
6. **Closure under scalar multiplication.** If λ is in $\mathbb{F}$ and $\vec{v}$ is in V , then $\lambda\vec{v}$ is in V .
7. **"Associativity" of multiplication.** For all $\lambda, \mu \in \mathbb{F}$ and all $\vec{v} \in V$ we have $\lambda(\mu\vec{v}) = (\lambda\mu)\vec{v}$.
8. **Multiplication by (the scalar) 1.** For each $\vec{v} \in V$ we have $1\vec{v} = \vec{v}$.
9. **Scalar distributive law.** For all λ, μ in $\mathbb{F}$ and all $\vec{v}$ in V we have $(\lambda + \mu)\vec{v} = \lambda\vec{v} + \mu\vec{v}$.
10. **Vector distributive law.** For all λ in $\mathbb{F}$ and all $\vec{u}, \vec{v}$ in V we have $\lambda(\vec{u} + \vec{v}) = \lambda\vec{u} + \lambda\vec{v}$.

Warm up: Burning questions from last time

Thanks to the students who came at the end of last lectures and asked those questions!

Warm up. True-False-Salvage

a) True or false?

- (i) The real polynomials of degree 3 form a vector space over $\mathbb{R}$.
- (ii) The real polynomials of degree 17 form a vector space over $\mathbb{R}$.
- (iii) Let n be a positive integer. The real polynomials of degree n form a vector space over $\mathbb{R}$.



b) Modify the above statements which are false, if any, to make them true.





UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 2: Properties of Vector Arithmetic

Lecturer: Aidan Sims (aidan.sims@unsw.edu.au)

Slides: David Angell, Laure Helme-Guizon and many others

More properties true in all vector spaces

By doing a couple of sample calculations, we have identified ten important properties of vectors and have used them as the definition of a vector space. But there are many more important algebraic facts about vectors that you have learned and used in MATH1131, for example,

$$0\vec{v} = \vec{0} \quad -(\vec{u} + \vec{v}) = (-\vec{u}) + (-\vec{v}) \quad -(-\vec{v}) = \vec{v} .$$



Should we include these properties in the vector space axioms?

We do not include these among the vector space axioms because it turns out that it is not necessary – we can *prove* they are true by using the axioms. Once we have done this for a general vector space, we know that the properties are automatically true for *all* vector spaces, and we never need to prove them again!

It would be possible to include extra axioms so that these (or other) properties are true *by definition* in a vector space. The drawback to this is that then it would be harder to prove that something actually is a vector space! We already have to check ten things if we want to know that a set is a vector space: ten is quite enough without giving ourselves even more work.

Fact (Properties of vector addition)

In any vector space V , the following properties hold.

1. **Uniqueness of zero.** A vector space has *only one* zero vector.
2. **Cancellation.** For any vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ in V ,
if $\vec{u} + \vec{v} = \vec{u} + \vec{w}$ then $\vec{v} = \vec{w}$.
3. **Uniqueness of negatives.** Each vector in a vector space has only one negative.
4. **Negative of the zero vector.** We have $-\vec{0} = \vec{0}$.

Proof of (1.) To show that V has only one zero vector, suppose that vectors $\vec{0}$ and $\vec{0}'$ are both zero vectors. Then

$$\begin{aligned}\vec{0} &= \vec{0} + \vec{0}' && \text{since } \vec{0}' \text{ is a zero vector} \\ &= \vec{0}' + \vec{0} && \text{by the commutative law} \\ &= \vec{0}' && \text{since } \vec{0} \text{ is a zero vector.}\end{aligned}$$

Therefore $\vec{0}$ and $\vec{0}'$ are in fact the same, and so V contains only one zero vector.

Properties of vector addition, continued

Exercise 1. Prove (3): Each vector in a vector space has only one negative.



Exercise 2. Prove (4): The zero vector is its own negative, i.e. $-\vec{0} = \vec{0}$.



Comment. None of these four results should be surprising, or difficult to remember!

Question. So why bother proving them?

Fact (Properties of scalar multiplication)

In any vector space V , the following properties hold.

1. **Multiplication by the zero scalar.** For every $\vec{v}$ in V we have $0\vec{v} = \vec{0}$.
2. **Multiplication by the zero vector.** For every scalar λ we have $\lambda\vec{0} = \vec{0}$.
3. **Multiplication by -1 .** For every $\vec{v}$ in V we have $(-1)\vec{v} = -\vec{v}$.
4. **Zero products.** Let $\vec{v} \in V$ and let λ be a scalar.
If $\lambda\vec{v} = \vec{0}$ then either $\lambda = 0$ or $\vec{v} = \vec{0}$.
5. **Multiplicative cancellation.** Let $\vec{v} \in V$ and let λ, μ be scalars.
If $\lambda\vec{v} = \mu\vec{v}$ and $\vec{v} \neq \vec{0}$ then $\lambda = \mu$.

Comments.

- Once again, these properties should be intuitively obvious, and easy to remember.
- In regard to property 3, note that $(-1)\vec{v}$ and $-\vec{v}$ are *conceptually* very different: the former is the result of multiplying $\vec{v}$ by a certain scalar, while the latter is a vector which produces $\vec{0}$ when added to $\vec{v}$. So perhaps it is not so obvious, after all, that they are equal!
- What is wrong with the following attempted proof for the first part?
“Let $\vec{v} = (x_1, x_2, \dots, x_n)$. Then $0\vec{v} = (0x_1, 0x_2, \dots, 0x_n) = (0, 0, \dots, 0) = \vec{0}$.”

Proof

We shall prove the first part of the previous result, and shall leave the others as exercises.

Proposition. *Multiplication by the zero scalar.* Let V be a vector space. Then for every $\vec{v}$ in V we have $0\vec{v} = \vec{0}$.

Proof. Let $\vec{v} \in V$. Then

$$\begin{aligned} 0\vec{v} + 0\vec{v} &= (0 + 0)\vec{v} && \text{(scalar distributive law)} \\ &= 0\vec{v} \\ &= 0\vec{v} + \vec{0} && \text{(zero vector)} \end{aligned}$$

Thus $0\vec{v} + 0\vec{v} = 0\vec{v} + \vec{0}$, and by the *cancellation property* we have $0\vec{v} = \vec{0}$. This is what we wanted to prove.

Proof advice

Setting out of the problem is very important!

- Write “Proposition” or “Theorem” or “Lemma”, and then *carefully and accurately* state what you intend to prove.
- On a new line write the heading “Proof”; then give your argument.
- Give a reason for every step.
- Write clearly! Use complete sentences, including capital letters, full stops and other punctuation.

Even more important is the idea / structure of the proof

Good news, there are general patterns, let's go through them.

How to prove something is TRUE?

- To prove a statement of the form “**There exists** $x \in D$ such that $P(x)$ is true”, the proof should follow the structure:

Idea: “Of course there exists one value for which it is true because, see, here is one value which works.”

Choose $x \in D$ to be ...

$\vdots$

Then $P(x)$ is true.

- To prove a statement of the form “**For all** $x \in D$, $P(x)$ is true”, the proof should follow the structure:

Idea: “Since no matter which value I pick, it is true for that one, then it must be true for all of them.”

Let $x \in D$. OR **Suppose** $x \in D$.

$\vdots$

Then $P(x)$ is true.

Exercise 3. Show that the vector space $\mathbb{R}^2$ under the usual vector addition and scalar multiplication is a vector space. (That is, show that $\mathbb{R}^2$ satisfies all ten vector space axioms.)

How to prove something is FALSE?

- To **disprove** a statement of the form “**For all** $x \in D$, $P(x)$ is true”, the proof should follow the structure:

Idea: “It cannot possibly be true for all of them: Look, I found one for which it is not true.”

Choose $x \in D$ to be ...

⋮

Then $P(x)$ is false.

- To **disprove** a statement of the form “**There exists** $x \in D$ such that $P(x)$ is true”, the proof should follow the structure:

Idea: “There cannot possibly be one value for which it is true because, see, no matter which one I pick, it is not true for that one.”

Let $x \in D$. OR Suppose $x \in D$.

⋮

Then $P(x)$ is false.

TIP!

Usually if we want to prove that a particular vector space axiom does not hold, we will be providing a disproof of the first form. This sort of argument is also known as **proof by counterexample**.